

# Federico P. Cortese, PhD

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Date of birth: January 1, 1994

[Personal Website](#) - [Google Scholar](#) - [GitHub](#) - [ORCID](#) - [ResearchGate](#)

## RESEARCH INTERESTS

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Latent variable models, multivariate time-series analysis, regime switching models, unsupervised learning

## PROFESSIONAL POSITIONS

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### Research fellow

Milan, Italy

*Department of Economics, Management, and Quantitative Methods – University of Milan*

*July 2025 – Present*

*Supervisor: Prof. Luca Rossini - Research: 100%, Teaching: optional*

### Researcher

Milan, Italy

*National Research Council, Institute for Applied Mathematics and Information Technologies*

*October 2023 – June 2025*

*Supervisor: Dr. Antonio Pievatolo*

### Visiting research fellow

Lund, Sweden

*Centre for Mathematical Sciences, Lund University - Supervisor: Prof. Erik Lindström*

*May 2022 – October 2022*

## EDUCATION

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### PhD in Statistics

Milan, Italy

*University of Milano-Bicocca - Supervisor: Prof. Fulvia Pennoni*

*2019 – 2023*

Thesis: [“Statistical Modeling and Temporal Clustering of Multivariate Time-Series with Applications to Financial Data”](#)

### MSc in Finance and Insurance

Rende, Italy

*University of Calabria - Supervisor: Prof. Arturo Leccadito*

*2017 – 2019*

Thesis: “Tail Dependence in Financial Markets: A Dynamic Copula Approach”

Final grade: 110/110 cum laude

### BSc in Mathematics

Rende, Italy

*University of Calabria - Supervisor: Prof. Vittorio Colao*

*2013 – 2017*

Thesis: “Stochastic Differential Equations with Financial Applications” (in italian)

Final grade: 100/110

## ARTICLES

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Cortese, Federico P.; Rossini, Luca

2026

[A comparison between initialization strategies for the infinite hidden Markov model](#)

*Computational Statistics & Data Analysis (CSDA)*

Cortese, Federico P.; Pievatolo, Antonio

2026

[Spatio-temporal jump model for urban thermal comfort monitoring](#)

*Stochastic Environmental Research and Risk Assessment (SERRA)*

Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik

2026

[Generalized information criteria for high-dimensional sparse statistical jump models](#)

*AStA: Advances in Statistical Analysis*

Algieri, Bernardina; Cortese, Federico P.; Lawuobahsumo, Kokulo; Leccadito, Arturo

2026

[The dance of the markets: Unveiling Bitcoin’s time-varying financial correlations using a GAS-based approach](#)

*China Finance Review International*

Cortese, Federico P.; Di Ruzza, Sara; Alessi, Elisa Maria

2025

[A statistical sparse jump model for automatic identification of dynamical transitions in the co-orbital regime](#)

*Nonlinear Dynamics*

Cortese, Federico P.; Pievatolo, Antonio

2025

[Statistical jump model for mixed-type data with missing data imputation](#)

*Advances in Data Analysis and Classification*

Cortese, Federico P.; Pennoni, Fulvia; Bartolucci, Francesco 2024

[Maximum likelihood estimation of regime switching Student-\*t\* copula model](#)

*International Statistical Review*

Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik 2023

[What drives cryptocurrency returns? A sparse statistical jump model approach](#)

*Digital Finance*

Cortese, Federico P. 2019

[Tail dependence in financial markets: A dynamic copula approach](#)

*Risks*

## SHORT PAPERS

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Cortese, Federico P.; Pievatolo, Antonio 2025

[Thermal Comfort Monitoring of Urban Areas: A Spatio-Temporal Jump Model Approach](#)

*Innovation & Society - Statistics and Data Science for Evaluation and Quality*

Cortese, Federico P.; Pievatolo, Antonio; Alessi, Elisa Maria 2025

[Fuzzy Jump Model for Asteroids Co-orbital Regimes Identification](#)

*Statistics for Innovation II - Springer Proceedings in Mathematics & Statistics*

Cortese, Federico P.; Pievatolo, Antonio 2024

[Modelling and forecasting correlated failure counts](#)

*ICES 2024 - 2nd Italian Conference on Economic Statistics - Book of Short Papers*

Cortese, Federico P.; Kolm, Petter N.; Lindström, Erik 2023

[Generalized information criteria for sparse statistical jump models](#)

*Symposium i anvendt statistik - Copenhagen Business School*

Cortese, Federico P.; Bartolucci, Francesco; Pennoni, Fulvia 2022

[A regime switching Student-\*t\* copula model for the analysis of cryptocurrencies data](#)

*Book of Abstracts, Mathematical and Statistical Methods for Actuarial Sciences and Finance*

Bartolucci, Francesco; Pennoni, Fulvia; Cortese, Federico P. 2021

[Hidden Markov and regime switching copula models for state allocation in multiple time-series](#)

*Book of abstracts and short papers, CLADAG, Firenze University Press*

## PREPRINTS

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Cortese, Federico P.; Farcomeni, Alessio 2026

[Robust State-Conditional Feature-Weighted Jump Models for Temporal Clustering](#)

Cortese, Federico P.; Pievatolo, Antonio; Alessi, Elisa Maria 2025

[Fuzzy Jump Models for Soft and Hard Clustering of Multivariate Time Series Data](#)

## ONGOING PROJECTS

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- Leisen, Fabrizio; Cortese, Federico P.; Rossini, Luca  
*Copula-Based Compound Random Measures for the Modeling of Sparse Networks with Overlapping Community Structure*
- Cortese, Federico P.; Rossini, Luca  
*Multivariate Quantile Regression through Infinite Hidden Markov Models*
- Cortese, Federico P.; Alessi, Elisa Maria  
*rJM: An R Package for the Estimation of Statistical Jump Models of Various Types*

## TEACHING EXPERIENCE

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- Academic year **2025-2026**  
Lecturer – Statistics for Big Data for Economics and Business  
*BSc Economics and Management, University of Milan - 20 hours*  
Teaching assistant – Unsupervised Learning  
*MSc Data Science for Public Decision Making, University of Rome-Tor Vergata - 18 hours*
- Academic year **2024-2025**  
Teaching assistant – Unsupervised Learning  
*MSc Data Science for Public Decision Making, University of Rome-Tor Vergata - 12 hours*
- Academic year **2022-2023**  
Teaching assistant – Probability and Statistics  
*BSc Automation Engineering, Politecnico di Milano - 36 hours*  
Teaching assistant – Fundamentals of Statistics and Biomedical Signals  
*BSc Biomedical Engineering, Politecnico di Milano- 20 hours*  
Teaching assistant – Bayesian Inference  
*MSc Biostatistics, University of Milano-Bicocca- 12 hours*  
Tutor – Probability and Statistics for Computer Science  
*BSc Computer Science, University of Milano-Bicocca - 28 hours*
- Academic year **2021-2022**  
Teaching assistant – Fundamentals of Statistics and Biomedical Signals  
*BSc Biomedical Engineering, Politecnico di Milano- 20 hours*  
Teaching assistant – Bayesian Inference  
*MSc Biostatistics, University of Milano-Bicocca- 12 hours*
- Academic year **2018-2019**  
Tutor – Financial Econometrics  
*MSc Finance and Insurance, University of Calabria- 10 hours*

## THESIS SUPERVISION

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- Simone Bosio - *University of Brescia - MSc in Economics* (co-supervised with prof. Enrico Ripamonti)

## SOFTWARE

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- [RSstcopula](#)  
R/C++ implementation for estimation and simulation of regime-switching Student- $t$  copula models.
- [GIC-for-SJM](#)  
R/C++/Python implementation for estimation of Sparse Jump Models and model selection via Generalized Information Criteria.

## RESEARCH PROJECTS

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[LUMIO](#). Development of data analysis and machine-learning models to study asteroid dynamics and predict orbital evolution for the LUMIO mission.

[RAISE](#). Implementation of statistical models and data analysis strategies for urban thermal comfort within the Robotics and AI for Socio-economic Empowerment project, supported by the European Commission under NextGenerationEU.

URBANSENSE. Supervised and monitoring of the activities related the project – “Citizens as sensors: promoting inclusion through the development and application of a multifunctional wearable system for a citizen-oriented city”.

## CONFERENCES

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- ENBIS-26, 26th annual conference of the European Network for Business and Industrial Statistics - University of Florence - **Invited speaker**, September 7-9, 2026
- MBC<sup>2</sup> 2026, 8th Workshop on Models and Learning in Clustering and Classification, University of Catanaia - **Contributed Speaker**, August 25-28, 2026
- SIS-FENStatS 2026, 53rd SIS and 1st FENStatS Scientific Meetings, La Sapienza - University of Rome, - **Contributed Speaker**, June 22-25, 2026
- 5th ISI Regional Statistics Conference - Valletta, University of Malta - **Contributed Speaker**, June 3-5 2026
- CMStatistics2025, 19th International Conference of the ERCIM Working Group on Computational and Methodological Statistics and the network Computational and Financial Econometrics - Birkbeck, University of London - **Invited Speaker**, December 13-15 2025
- ISBIS 2025, International Society for Business and Industrial Statistics - University of Amsterdam - **Invited Speaker**, October 2-3 2025
- EPSC-DPS 2025, Europlanet Science Congress, Division for Planetary Sciences - Finlandia Hall, Helsinki - **Contributed Speaker**, September 7-12, 2025
- IES 2025, Innovation & Society: Statistics and Data Science for Evaluation and Quality - University of Padua, Brixen - **Invited Speaker** and **discussant**, June 25-27, 2025
- SIS 2025, Statistics for Innovation - University of Genoa - **Contributed Speaker**, June 16-18, 2025
- BISP14, Bayesian Inference in Stochastic Processes - National Research Council of Italy, Institute for Applied Mathematics and Information Technologies, Milan - **Chair**, May 26-28, 2025
- SDS, Statistics and Data Science - University of Milan and Politecnico di Milano, Italy - **Invited Speaker**, April 02-03, 2025
- Emerging Risks in Economics and Finance - University of Florence, Italy - **Invited Speaker**, September 12-13, 2024
- EcoSta 2023 - Waseda University, Tokyo, Japan - **Contributed Speaker**, August 1-3, 2023
- Statistics@Naples - University of Naples Federico II, Naples, Italy - **Contributed Speaker**, June 28-30, 2023
- CMStatistics2022, Computational and Methodological Statistics - King's College London, London, UK - **Invited Speaker**, December 17-19, 2022
- International FinTech Research Conference - Politecnico di Milano, Milan, Italy - **Contributed Speaker**, October 27-28, 2022
- MAF2022, Mathematical and Statistical Methods for Actuarial Sciences and Finance - University of Salerno, Fisciano (SA), Italy - **Contributed Speaker**, April 20-22, 2022
- CLADAG, 13th Scientific Meeting of the Classification and Data Analysis Group - University of Florence, Florence, Italy - **Participant**, September 9-11, 2021

## WORKSHOPS

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- “Advanced Time Series Methods” - University of Perugia, Perugia, Italy - Participant, June 19-20, 2023
- “Bayesian Nonparametrics for Complex Data” - University of Padua, Padua, Italy - Participant, January 24, 2020

## OUTREACH ACTIVITY

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*Escape Moon*. Educational activity developed within the [Co-Science](#) initiative to explain, through a game-based approach, the dynamics of celestial bodies and the Earth–Moon system to primary school children.

*Space debris: what future for a sustainable use of near-Earth space?* Public outreach event within the [Co-Science](#) initiative focused on the environmental challenges of space debris and the importance of sustainable practices in space exploration and satellite operations.

## EDITORIAL ACTIVITY

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Referee for *Annals of Operations Research* (Springer), *Statistics and Probability Letters* (Elsevier), *Nonlinear Dynamics* (Springer), *Statistical Modelling* (Sage Journals), *Metron* (Springer), *Discover Analytics* (Springer), *Advances in Computational Science and Engineering* (American Institute of Mathematical Sciences) and *Scientific Reports* (Nature).

## TECHNICAL SKILLS

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Proficient in R

Competent in Python

Familiar with MATLAB, C++, SAS

Softwares:  $\text{\LaTeX}$ , R-Markdown, Git

Languages: Italian (native), English (fluent), French (basic)